Supplementary Table 1. Detailed Definition of Gastroesophageal Reflux Disease

First author	Definition of GERD
Bayerdorffer[14]	Patients presenting to their general practitioner or gastrointestinal specialist with symptoms suggestive of GERD and with heartburn as their predominant symptom for longer than 6 months.
Bour[15]	Men and women aged ≥18 years were enrolled using the following inclusion criteria: presenting a relapse of their GERD symptoms, heartburn or acid regurgitation, whose intensity was at least moderate (Likert score > 2), having frequent symptomatic relapses (needing ≥ 2 treatments prescribed by his general practitioner in the 12 months prior to enrolment).
Cho[16]	The diagnosis of GERD was established by upper GI endoscopy and classified according to the modified Los Angeles classification grades (M, A, B, C, or D) before initial PPI treatment.
Hansen[17]	Male and female patients over 18 years of age with symptoms suggestive of GERD (heartburn as the predominant symptom with or without acid regurgitation) for 3 days or more in the 7 days before the start of the study were eligible for entry into the study.
Janssen[18]	Patients aged 18 to 75 years who had endoscopically confirmed Savary/Miller grade 0 or I GERD and who experienced frequent episodes of at least one of the characteristic GERD symptoms (heartburn, acid regurgitation, and/or pain on swallowing) during the previous 3 months were eligible for inclusion.
Morgan[19]	Male and female participants with a minimum 3-month history of GERD, with heartburn as the predominant symptom
Nagahara[20]	Eligible patients were those with a diagnosis of GERD by upper GI endoscopy of either the modified LA classification grade M, A, B, C, or D before the initial PPI treatment and who provided written informed consent to participate in this study.
Sjöstedt[21]	Patients with verified erosive reflux esophagitis (LA grades A-D), confirmed by endoscopy within 7 days prior to visit 1, were enrolled provided they fulfilled the following entry criteria. In addition to their verified esophagitis, they had to have a history of heartburn episodes over at least 6 months and at least 4 days with heartburn episodes during the week prior to visit 1.
Szucs[22]	Male and female patients (≥ 18 years of age) seeking primary care for symptoms suggestive of GERD (heartburn as the predominant symptom with or without acid regurgitation; for 3 days or more during the last 7 days), were enrolled into the study.
Tsai[23]	Patients, aged 18-80 years, with heartburn identified as their predominant symptom of GERD were eligible to enter the study if they had experienced heartburn on at least 4 days of the previous 7. In addition, patients were required to have a history of heartburn for 6 months or longer and no esophageal mucosal breaks as verified by endoscopy up to 14 days before enrolment into the study.
van der Velden[24]	Criteria for inclusion were: between 18 and 75 years of age, a history of GERD or GERD-like symptoms, no recent consultation for inadequately controlled symptoms.

GERD, gastroesophageal reflux disease; GI, gastrointestinal; PPI, proton pump inhibitor; LA, Los Angeles.

Supplementary Table 2. Details of Primary Outcomes of Included Studies

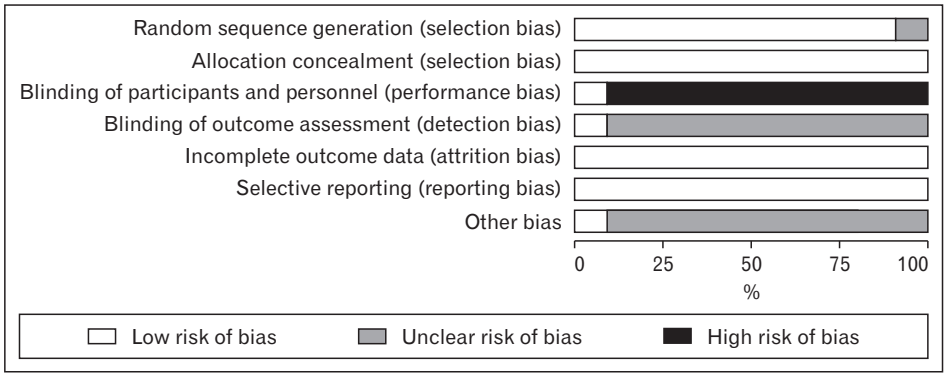
First author	Primary outcome	Details of primary outcome
Bayerdorffer[14]	Discontinuation due to unsatisfactory treatment	At clinical visits 2 to 5 the investigator confirmed with the patient if he/she wished to continue with the treatment and, if not, the date and reasons for discontinuation were recorded.
Bour[15]	Rate of symptom relief	The primary efficacy variable was symptom relief as assessed at 6 months. Symptom relief was defined as a Likert score of ≤ 2 for the symptom that had led to the initial consultation.
Cho[16]	Intensities and frequencies of heartburn	Alleviation of symptoms was defined as a Likert scale or frequency scale of 0 or 1 for the severity and frequency of heartburn or regurgitation.
Hansen[17]	Percentage of patients with no heartburn	Symptom severity at the 3-month and final visit was presented descriptively. Severity of GERD symptoms in each treatment strategy was further analysed at the 3-month and final visit by using the last value carried forward in case of discontinuation from the allocated strategy.
Janssen[18]	Rate of failure to control GERD symptoms	The primary variable in this study was the difference in the estimated rated of patients with uncontrolled heartburn, acid regurgitation, and pain on swallowing between the 2 treatment groups.
Morgan[19]	Proportion of heartburn-free days	The occurrence and severity of heartburn symptoms during the previous week was recorded using questionnaire. Severity was assessed as 0 = none; 1 = mild (present but causing little or no discomfort and can be ignored when not thought about); 2 = moderate (cannot be ignored but does not influence daily routine); 3 = severe (causes discomfort and some interference with daily routine); or 4 = very severe (disabling and interferes considerably with daily routine).
Nagahara[20]	Symptom relief	Patients' symptoms and QOL were assessed by the GOL scale and the QOLRAD-J. Our first aim was to compare the percentage of patients who achieved symptom relief at 4, 8, 16, and 24 weeks in each study group with relief from symptoms as the primary endpoint.
Sjöstedt[21]	Endoscopic relapse	The primary end-point of the study was the proportion of patients in endoscopic remission over time, up to 6 months.
Szucs[22]	Frequency of heartburn and regurgitation	These symptoms were classified as: none, mild (awareness of symptoms, but easily tolerated), moderate (discomfort sufficient to cause interference with normal activities), and severe (incapacitating, with inability to perform normal activities). Frequency of heartburn was recorded at the beginning and at the end of the initial symptom treatment course as the number of days with episodes during the last 7 days.
Tsai[23]	Discontinuation due to unwillingness to continue	The primary outcome measure was the time to discontinuation from the maintenance phase due to unwillingness to continue.
van der Velden[24]	The percentage of patients taking 6 or more tablets on-demand per week	Primary outcome was the volume of rescue medication per week. Three dosage groups were created: 0 to < 2 tablets/week ([almost] quitting treatment), 2 to < 6 /week (substantial reduction), 6-7/week (ongoing daily treatment).

GERD, gastroesophageal reflux disease; QOL, quality of life; GOS, Global Overall Symptom; QOLRAD-J, Japanese version of the Quality of Life in Reflux and Dyspepsia.

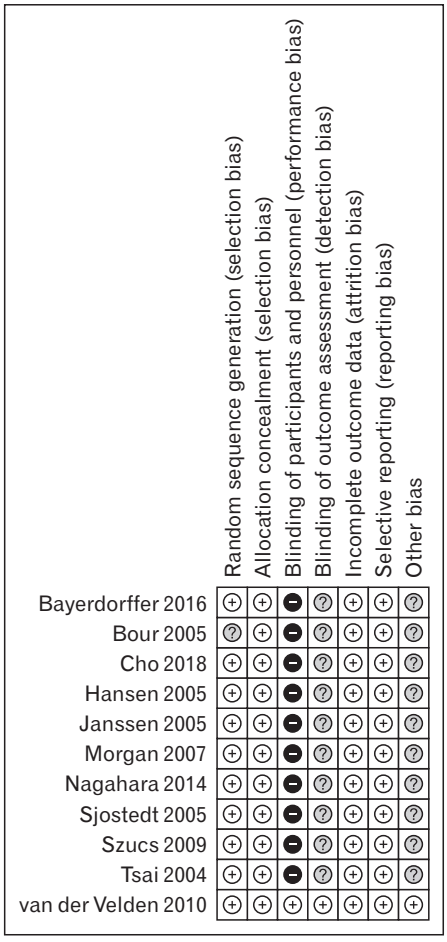
Supplementary Table 3. Sensitivity Analysis: The Meta-analysis Run Again, Systemically Omitting Each Study (Leave-One-Out Analysis)

Omitted study	Treatment failure RR (95% CI)	Symptom relief RR (95% CI)
Bayerdorffer et al[14]	1.40 (0.81, 2.41)	1.18 (1.04, 1.34)
Bour et al[15]		1.16 (1.04, 1.31)
Cho et al[16]		1.17 (1.04, 1.31)
Hansen et al[17]	1.23 (0.67, 2.24)	1.12 (1.04, 1.20)
Janssen et al[18]	1.35 (0.78, 2.34)	1.16 (1.03, 1.31)
Morgan et al[19]	1.21 (0.72, 2.04)	1.18 (1.04, 1.33)
Nagahara et al[20]		1.19 (1.07, 1.33)
Sjöstedt et al[21]	1.03 (0.67, 1.58)	1.14 (1.02, 1.27)
Szucs et al[22]	1.32 (0.71, 2.46)	1.17 (1.03, 1.34)
Tsai et al[23]	1.46 (0.90, 2.38)	
van der Velden et al[24]	1.13 (0.68, 1.90)	
Total (all studies included)	1.26 (0.76, 2.07)	1.16 (1.05, 1.29)

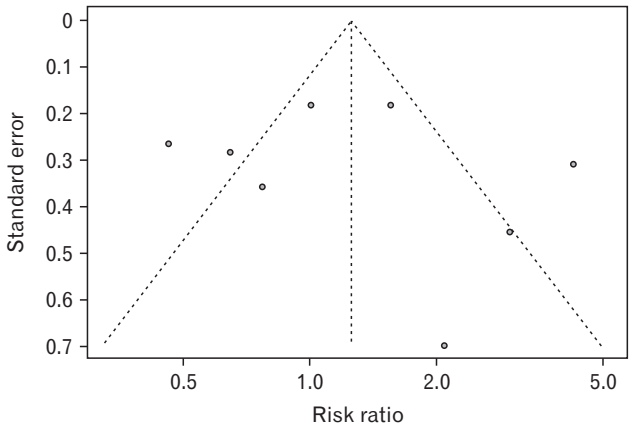
RR, risk ratio.



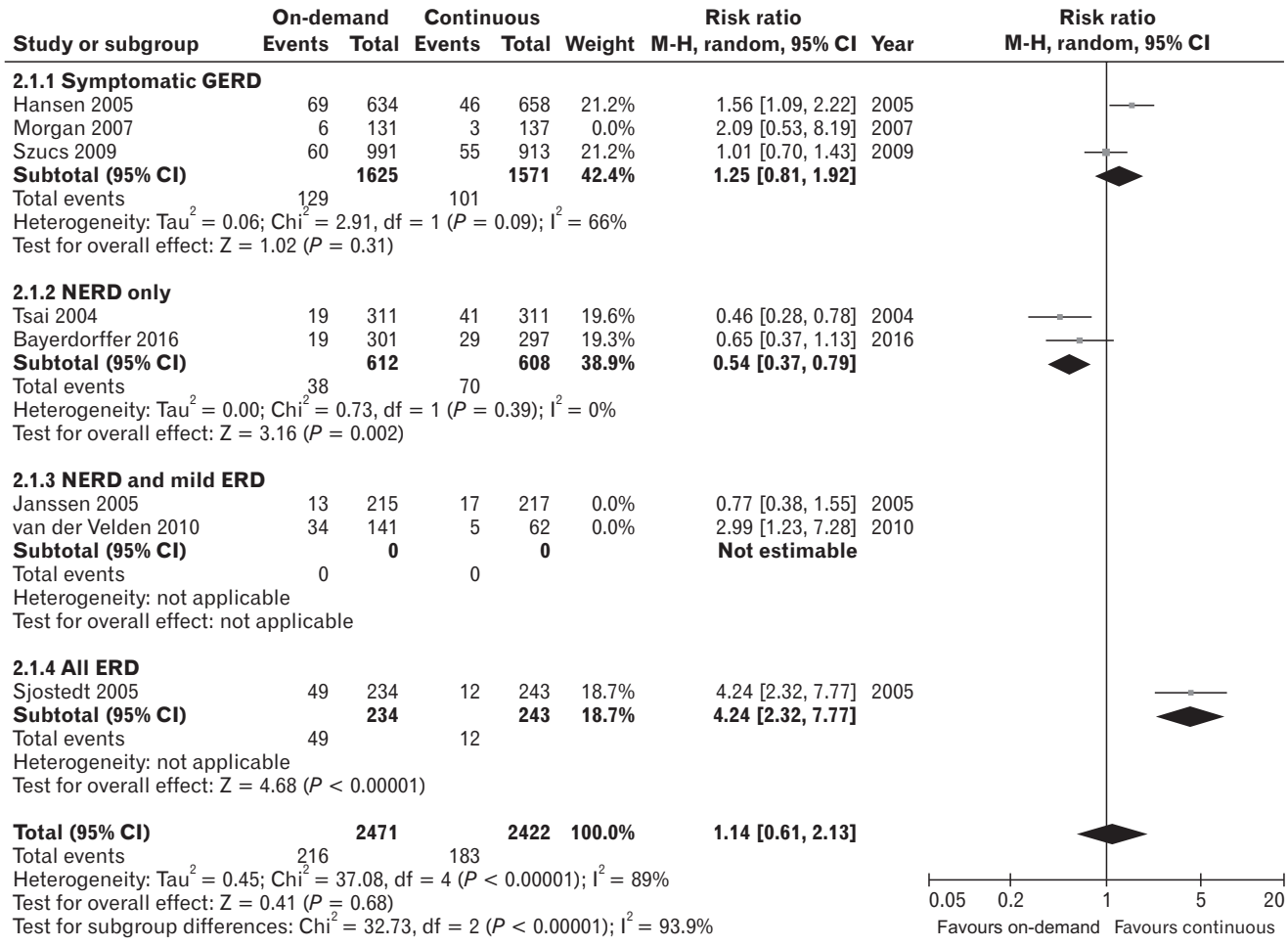
Supplementary Figure 1. Risk of bias graph of included studies.



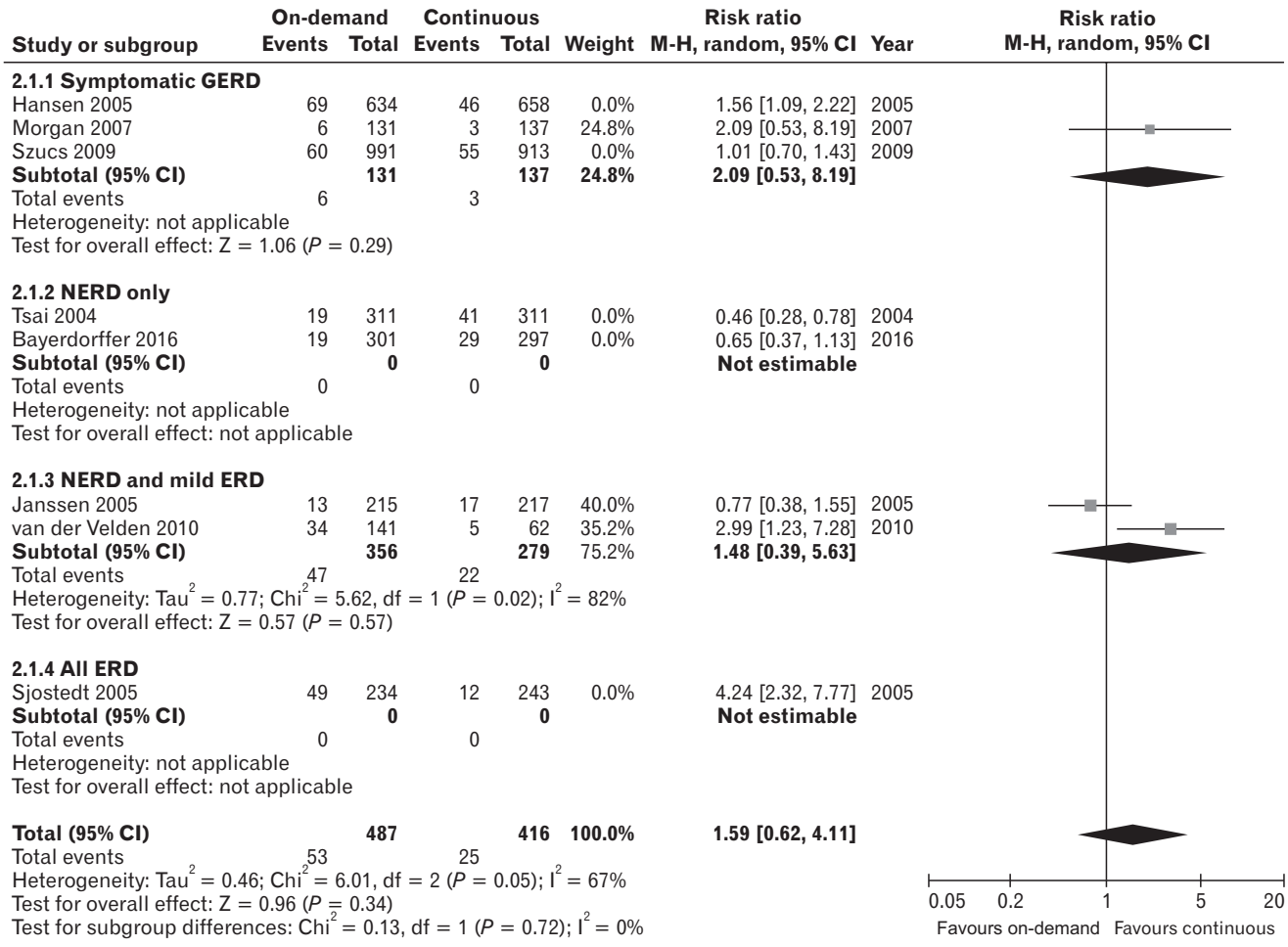
Supplementary Figure 2. Risk of bias summary of included studies.



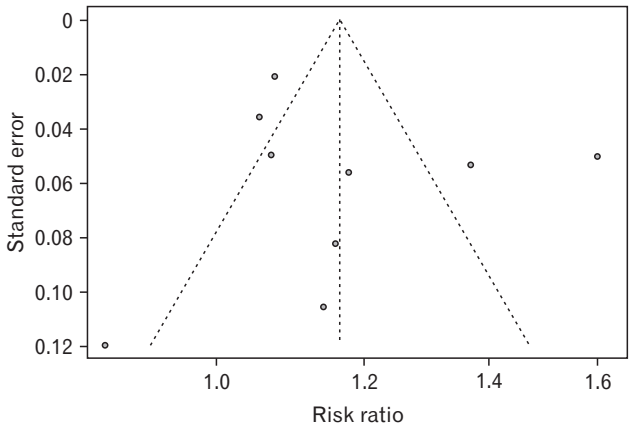
Supplementary Figure 3. Funnel plot for outcome of treatment failure of included studies.



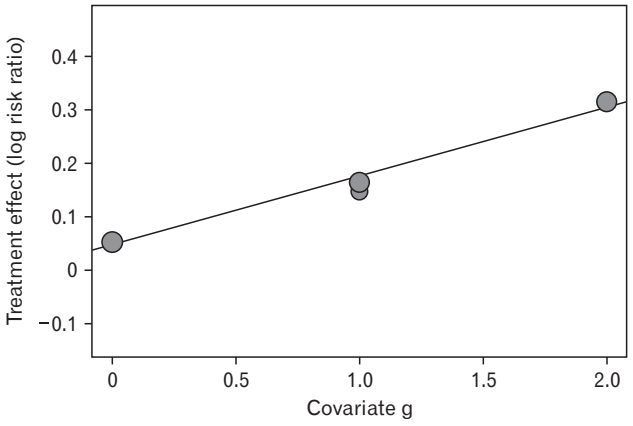
Supplementary Figure 4. Meta-analysis for treatment failure including studies using esomeprazole. M-H, Mantel-Haenszel; GERD, gastro-esophageal reflux disease; NERD, non-erosive esophageal reflux disease; ERD, erosive reflux disease.



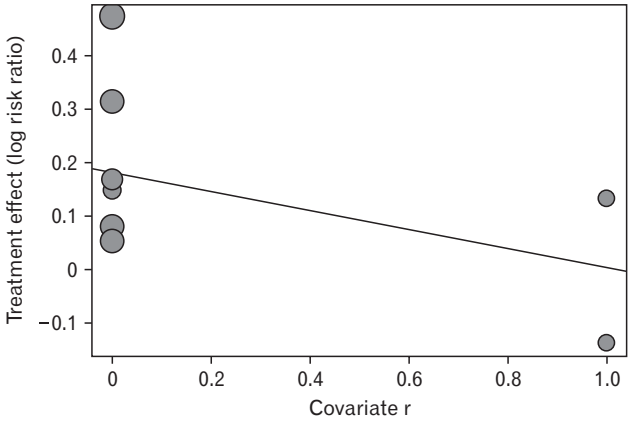
Supplementary Figure 5. Meta-analysis for treatment failure including studies using other proton pump inhibitors. M-H, Mantel-Haenszel; GERD, gastroesophageal reflux disease; NERD, non-erosive esophageal reflux disease; ERD, erosive reflux disease.



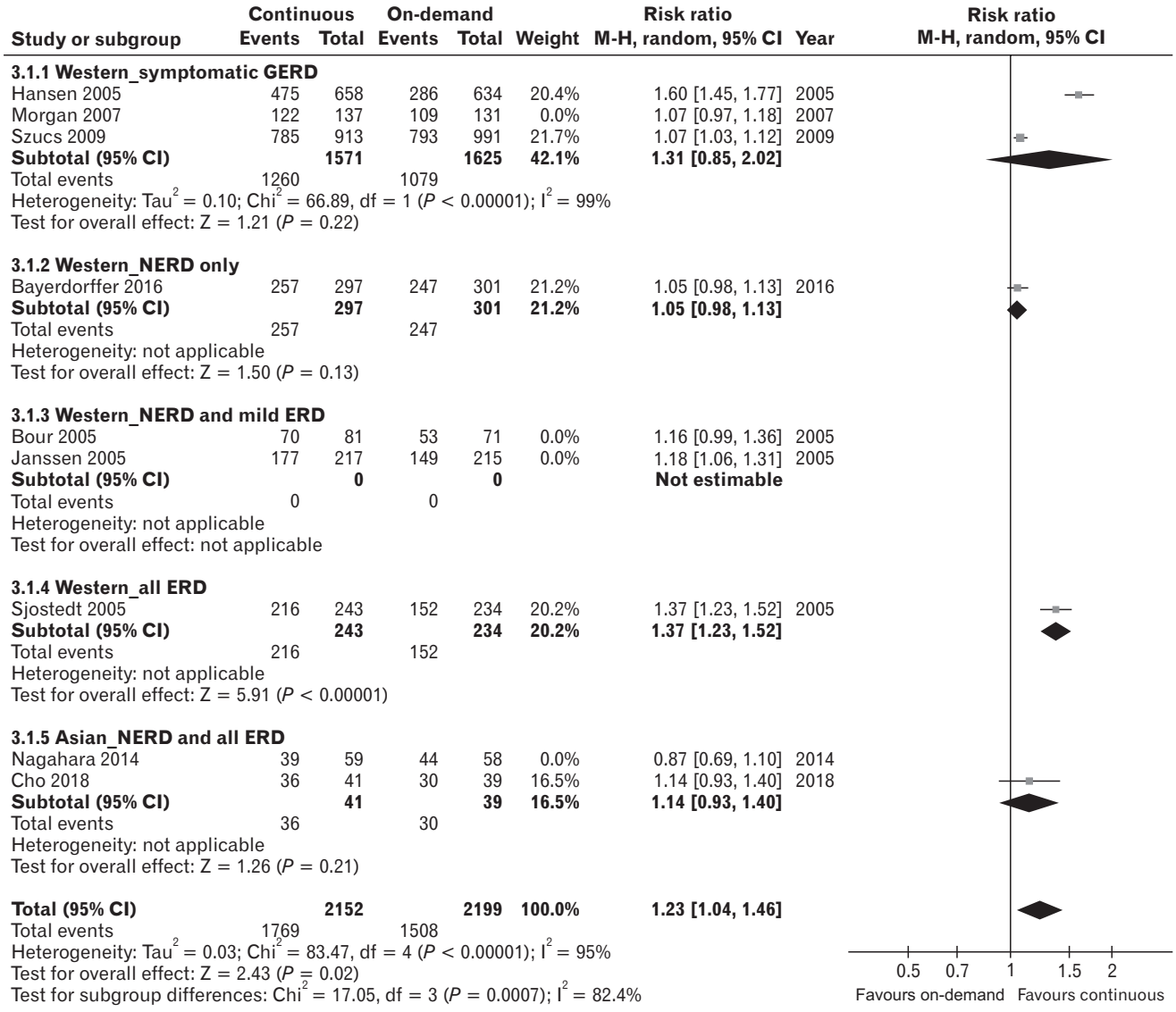
Supplementary Figure 6. Funnel plot for outcome of symptom relief of included studies.



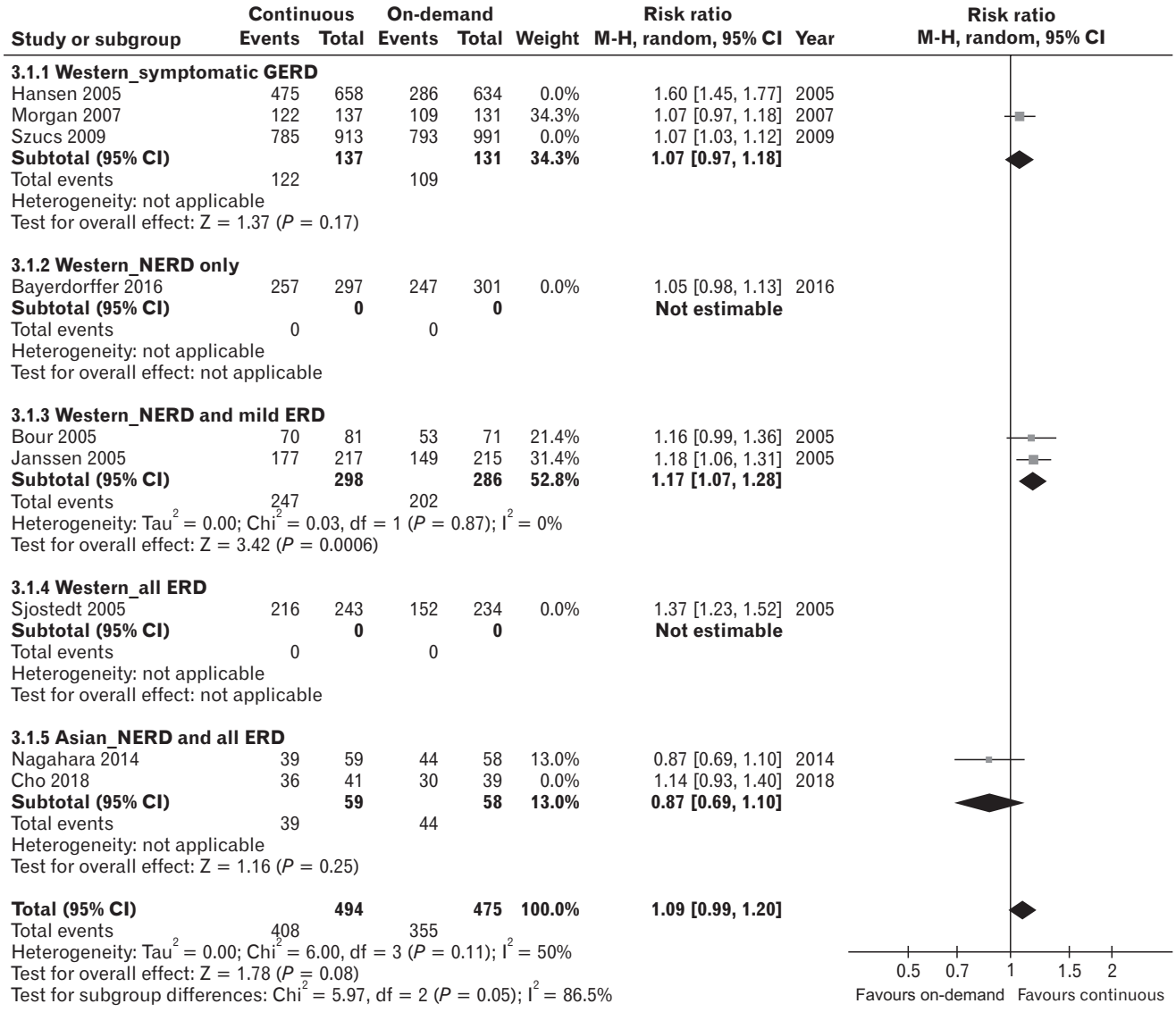
Supplementary Figure 7. Random-effects univariate meta-regression between symptom relief and esophagitis severity. Each circle represents a study and the size of the circle reflects the influence of that study on the model. The regression prediction is presented by the solid line (0.127 [95% CI, 0.066-0.188], $P < 0.001$).



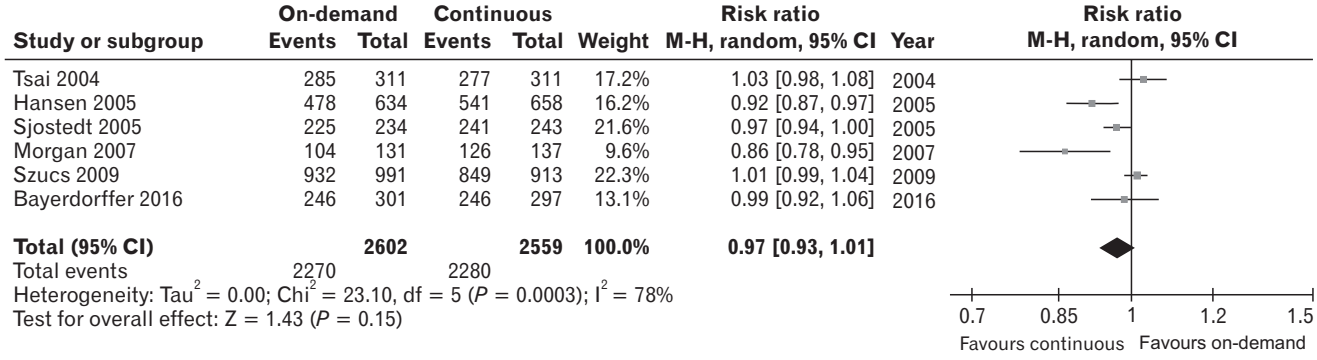
Supplementary Figure 8. Random-effects univariate meta-regression between symptom relief and region. Each circle represents a study and the size of the circle reflects the influence of that study on the model. The regression prediction is presented by the solid line (−0.179 [95% CI, −0.462-0.105], $P = 0.218$).



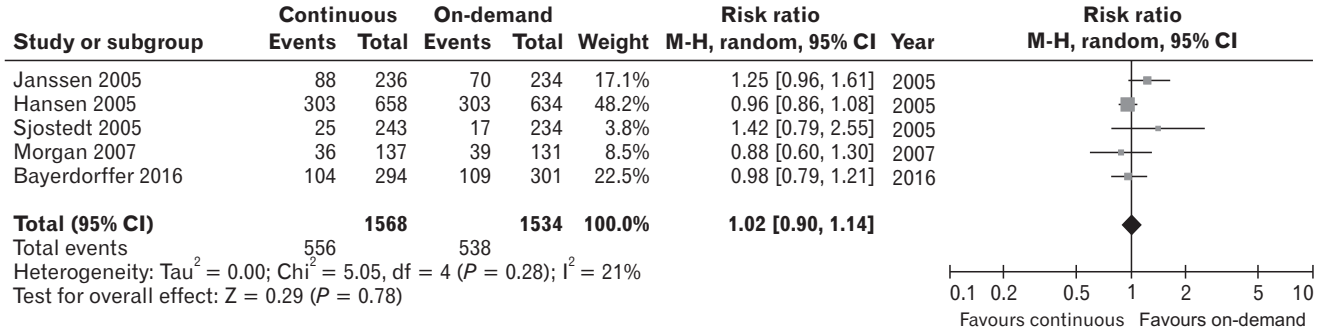
Supplementary Figure 9. Meta-analysis for symptom relief including studies using esomeprazole. M-H, Mantel-Haenszel; GERD, gastro-esophageal reflux disease; NERD, non-erosive esophageal reflux disease; ERD, erosive reflux disease.



Supplementary Figure 10. Meta-analysis for symptom relief including studies using other proton pump inhibitors. M-H, Mantel-Haenszel; GERD, gastroesophageal reflux disease; NERD, non-erosive esophageal reflux disease; ERD, erosive reflux disease.



Supplementary Figure 11. Forest plot of the relative efficacy in terms of the satisfaction between on-demand therapy and continuous therapy. M-H, Mantel-Haenszel.



Supplementary Figure 12. Forest plot of the adverse events in on-demand therapy group and continuous therapy group. M-H, Mantel-Haenszel.